1. From (15.58), we know that the number of virtual quanta per unit energy interval in the low frequency limit is

$$N(\hbar\omega) = \frac{2}{\pi}\left(\frac{e^2}{\hbar c}\right)\left(\frac{c}{v}\right)^2 \frac{1}{\hbar\omega}\left[\ln\left(\frac{1.123\gamma v}{\omega b_{\min}}\right) - \frac{v^2}{2c^2}\right] \tag{1}$$

Referring to the paragraph after Figure 15.8 (specifically with respect to electron disintegration), we set $b_{\min} = \hbar/\gamma m v$, $v \approx c$, and absorb $v^2/2c^2$ into the logarithm and write

$$N(\hbar\omega) = \frac{2}{\pi}\left(\frac{e^2}{\hbar c}\right)\frac{1}{\hbar\omega}\ln\left(\frac{k\gamma^2 mc^2}{\hbar\omega}\right) \qquad k \sim O(1) \tag{2}$$

Evidently, if $\sigma_{\text{photo}}(\omega)$ is the cross section for the photodisintegration of the nucleus by photon of frequency ω, the electrodisintegration cross section is the integral of $\sigma_{\text{photo}}(\omega)$ over applicable frequencies with the number of virtual quanta $N(\hbar\omega)$ as the weight, i.e.,

$$\sigma_{\text{el}}(E) = \int_{\omega_T}^{E/\hbar} N(\hbar\omega)\,\sigma_{\text{photo}}(\omega)\,d(\hbar\omega) = \frac{2}{\pi}\left(\frac{e^2}{\hbar c}\right)\int_{\omega_T}^{E/\hbar}\sigma_{\text{photo}}(\omega)\ln\left(\frac{k\gamma^2 mc^2}{\hbar\omega}\right)\frac{d\omega}{\omega} \tag{3}$$

2. With the resonance shape of σ_{photo},

$$\sigma_{\text{photo}}(\omega) = \frac{A}{2\pi}\frac{e^2}{Mc}\left[\frac{\Gamma}{(\omega-\omega_0)^2 + (\Gamma/2)^2}\right] \tag{4}$$

(3) becomes

$$\sigma_{\text{el}}(E) = \frac{2}{\pi}\left(\frac{e^2}{\hbar c}\right)\left(\frac{Ae^2}{Mc}\right)\int_{\omega_T}^{E/\hbar}\frac{1}{\pi}\left[\frac{\Gamma/2}{(\omega-\omega_0)^2+(\Gamma/2)^2}\right]\ln\left(\frac{kE^2}{mc^2\hbar\omega}\right)\frac{d\omega}{\omega} \tag{5}$$

Due to the relation

$$\delta(x) = \lim_{\epsilon\to 0^+}\frac{1}{\pi}\left(\frac{\epsilon}{x^2+\epsilon^2}\right) \tag{6}$$

when $\omega_0 - \omega_T \gg \Gamma$, we can take the approximation

$$\frac{1}{\pi}\left[\frac{\Gamma/2}{(\omega-\omega_0)^2+(\Gamma/2)^2}\right] \approx \delta(\omega-\omega_0) \tag{7}$$

The condition $E \gg \hbar\omega_0$ ensures the δ function has support within the integration range, thus putting (7) into (5) gives

$$\sigma_{\text{el}}(E) \approx \frac{2}{\pi}\left(\frac{e^2}{\hbar c}\right)\left(\frac{Ae^2}{Mc}\right)\frac{1}{\omega_0}\ln\left(\frac{kE^2}{mc^2\hbar\omega_0}\right) \tag{8}$$

To see the effect of finite resonance width, we can write (5) in normalized variables,

$$\eta = \frac{\omega}{\omega_0} \qquad x = \frac{E}{\hbar\omega_0} \qquad T = \frac{\Gamma}{\omega_0} \tag{9}$$

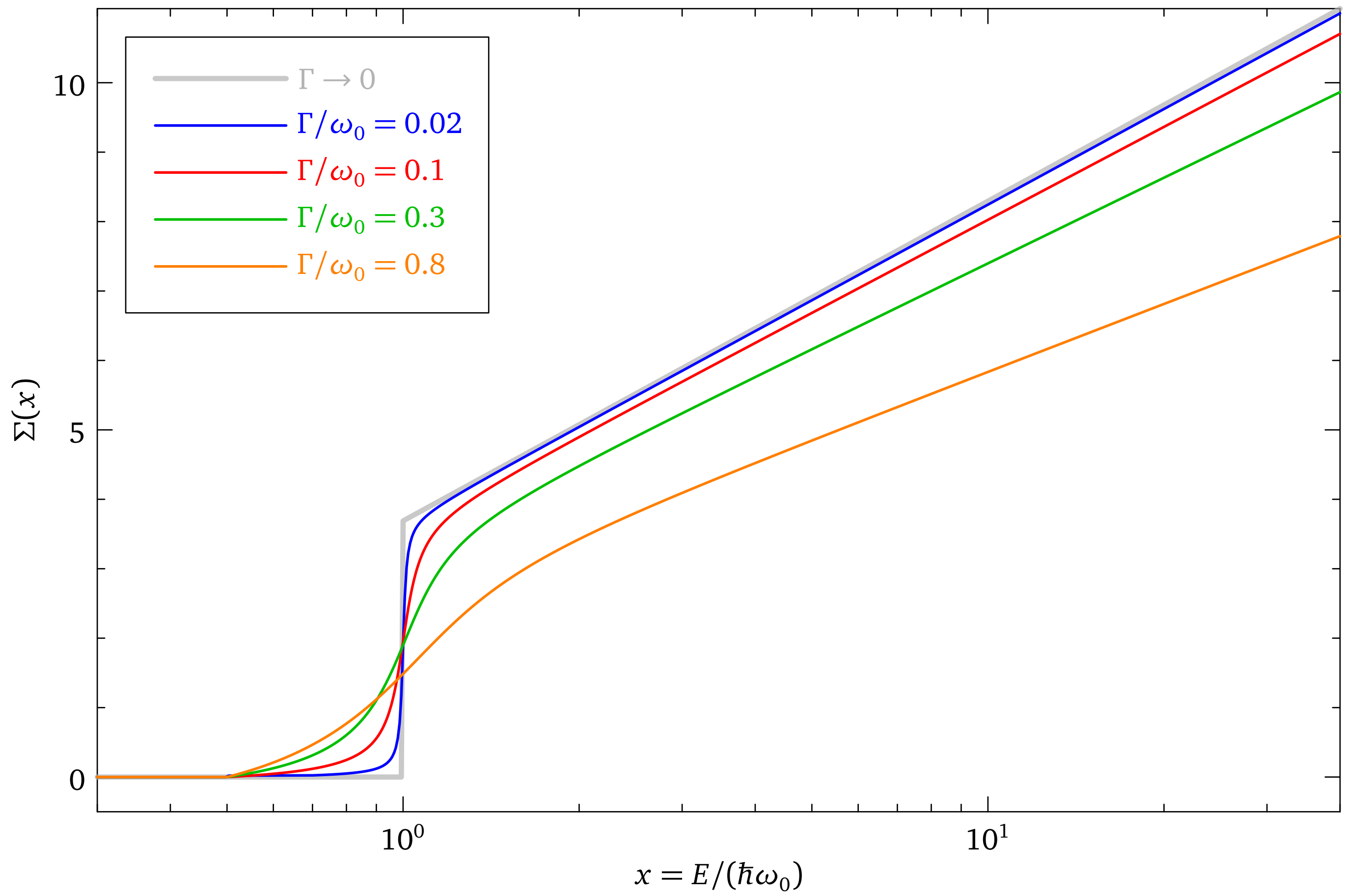
the electrodisintegration cross section can be written as

$$\sigma_{\text{el}}(x) = \frac{2}{\pi}\left(\frac{e^2}{\hbar c}\right)\left(\frac{Ae^2}{Mc}\right)\frac{1}{\omega_0}\int_{\omega_T/\omega_0}^{x}\frac{1}{\pi}\left[\frac{T/2}{(\eta-1)^2+(T/2)^2}\right]\ln\left[k\left(\frac{\hbar\omega_0}{mc^2}\right)\left(\frac{x^2}{\eta}\right)\right]\frac{d\eta}{\eta} \tag{10}$$

Setting $\omega_T = \omega_0/2$, $k\hbar\omega_0/mc^2 = 40$, we can plot the dimensionless integral

$$\Sigma(x) = \int_{1/2}^{x}\frac{1}{\pi}\left[\frac{T/2}{(\eta-1)^2+(T/2)^2}\right]\ln\left(\frac{40x^2}{\eta}\right)\frac{d\eta}{\eta} \tag{11}$$

with various $T = \Gamma/\omega_0$ values, as shown in the figure below.



3. The bremsstrahlung-induced cross section, per the problem's prescription, is (15.47) multiplied by $Ae^2/Mc\hbar\omega_0$:

$$\sigma_{\text{brem}} = \frac{d\chi}{d\omega} \cdot \frac{Ae^2}{Mc\hbar\omega_0} = \frac{16}{3}\frac{Z^2e^2}{c}\left(\frac{e^2}{mc^2}\right)^2 \ln\left(\frac{233}{Z^{1/3}}\right) \cdot \frac{Ae^2}{Mc\hbar\omega_0} \tag{12}$$

Dividing by $Z^2 r_0^2 = Z^2\left(e^2/mc^2\right)^2$ and forming the ratio with σ_{el} from (8):

$$F(E) = \frac{\sigma_{\text{brem}}/\left(Z^2 r_0^2\right)}{\sigma_{\text{el}}} = \frac{\frac{16}{3}\frac{e^2}{c} \cdot \ln\left(\frac{233}{Z^{1/3}}\right) \cdot \frac{Ae^2}{Mc\hbar\omega_0}}{\frac{2}{\pi}\frac{e^2}{\hbar c} \cdot \frac{Ae^2}{Mc\omega_0} \cdot \ln\left(\frac{kE^2}{mc^2\hbar\omega_0}\right)} = \frac{8\pi}{3} \cdot \frac{\ln\left(\frac{233}{Z^{1/3}}\right)}{\ln\left(\frac{kE^2}{mc^2\hbar\omega_0}\right)} \tag{13}$$

For $\hbar\omega_0 \approx 20$ MeV, noting that Z in (15.47) is the radiator's atomic number (aluminum, $Z = 13$ in Wolynec et al.):

$$F(E) \approx \frac{38.5}{\ln\left(\frac{kE^2}{mc^2\hbar\omega_0}\right)} \approx \begin{cases} 8.6 & E = 30 \text{ MeV} \\ 7.6 & E = 40 \text{ MeV} \end{cases} \tag{14}$$

These values are in good agreement with the data in Figures 1–5 of Wolynec et al., where $F \approx 8$–10 and decreases slowly with energy. The $1/\ln E^2$ energy dependence also matches the observed slow decline at the high-energy end.